

SingleRAN

MACsec Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-04-27).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for MACsec by UMPThb series main control boards. For details, see 4.3.3 Hardware .	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

3 Overview

3.1 Background

For BBU interconnection:

To deploy BBU interconnection-related features, UMPT boards must be used to directly interconnect BBUs through Ethernet ports (CI ports). Data is transmitted through these interconnection ports complying with the standard Ethernet protocols.

Ethernet is characterized by simplicity, high efficiency, and scalability. However, it provides simple packet formats that can be easily parsed and therefore is not secure. Data transmitted over the interconnection ports contains control-, user-, and management-plane information, which is all sensitive. Media Access Control security (MACsec) is required to protect the data.

For Enhanced Common Public Radio Interface (eCPRI):

The eCPRI protocol is used for the connection between a BBU and an AAU in a radio base station. Based on this protocol, a link is set up between the BBU and AAU to transmit data. Data transmitted over the link includes control-, user-, and synchronization-plane data. MACsec is required to protect the data, and it applies to NR FDD, NR TDD, LTE FDD, LTE TDD, and UMTS.

3.2 Introduction

MACsec defines secure data communication based on IEEE 802 LAN and complies with the IEEE Std 802.1AE-2006 protocol.

MACsec requires two communication peers (also known as MACsec peers) to encrypt packets and authenticate the data origin to ensure the following:

- Access device validity: MACsec provides the identity authentication function to ensure that only authorized devices can access a network.
- Data confidentiality: The MACsec entity encrypts data and transmits data in ciphertext to prevent unauthorized access to data.
- Data integrity: The MACsec entity checks integrity of the received data. Complete data indicates that the data has not been tampered with.

- Data authenticity: The MACsec entity authenticates the data origin to confirm the sender of the data.

 NOTE

The MACsec entity is an NE or network device that uses MACsec for communication.

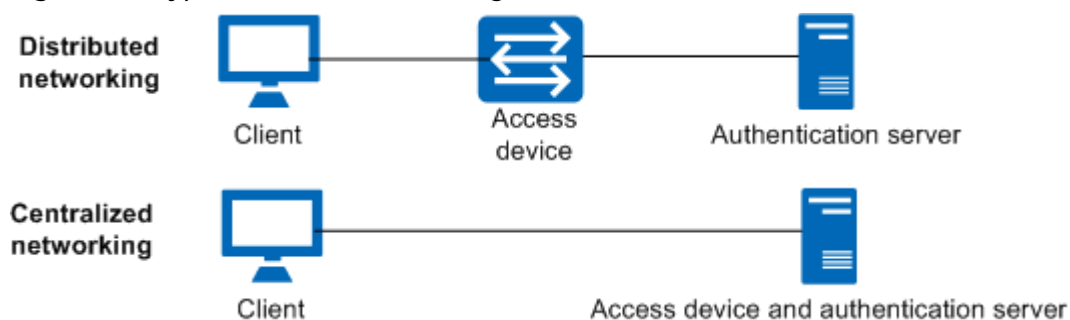
3.3 Architecture

MACsec involves the following devices: client, access device, and authentication server.

- The client refers to a terminal requesting access to a LAN. The access device in the LAN authenticates this terminal and performs the MACsec key negotiation and packet encryption functions.
- The access device controls access of clients. It interacts with the authentication server and performs 802.1x-based authentication on connected clients. In addition, the access device performs the MACsec key negotiation and packet encryption functions.
- The authentication server is generally a Remote Authentication Dial In User Service (RADIUS) server and it authenticates and authorizes clients. After a client is authenticated, the authentication server distributes keys to the client and the access device.

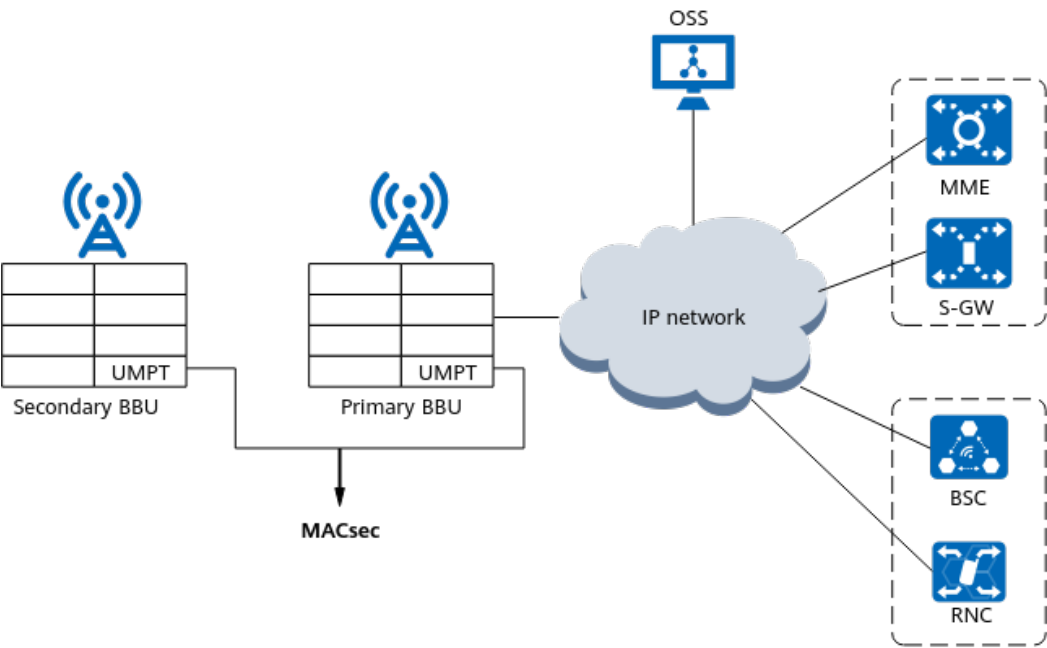
MACsec has two typical networking modes: distributed and centralized (as shown in [Figure 3-1](#)). Unlike distributed networking, the functions of the access device and authentication server are integrated into one physical device for information exchange in centralized networking.

Figure 3-1 Typical MACsec networking modes



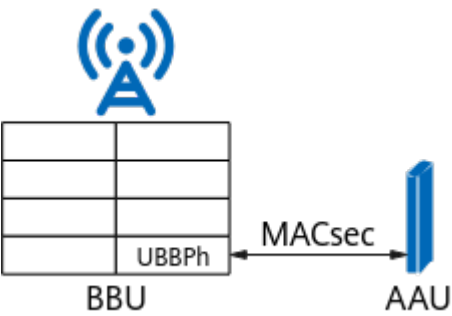
When BBUs are directly connected using a network cable, MACsec is deployed in centralized networking. The UMPT in the primary BBU serves as the access device and authentication server. The UMPT in the secondary BBU serves as the client. [Figure 3-2](#) shows the networking.

Figure 3-2 MACsec networking



When a BBU and an AAU are connected using the eCPRI protocol, MACsec is deployed in centralized networking. As shown in [Figure 3-3](#), the UBBPh in the BBU serves as the access device and authentication server, and the AAU serves as the client. For details about the board requirements, see [4.3.3 Hardware](#).

Figure 3-3 eCPRI MACsec networking



4 MACsec

4.1 Principles

4.1.1 Basic Concepts

MACsec is used with access control based on 802.1x. After access control based on 802.1x is successful, a MACsec procedure starts. After MACsec identifies packets sent by authenticated clients, MACsec uses the key, which is obtained during negotiation using MACsec Key Agreement (MKA), to encrypt and perform integrity check on authenticated user data. This prevents interconnection ports from processing packets sent from or tampered with by unauthenticated devices.

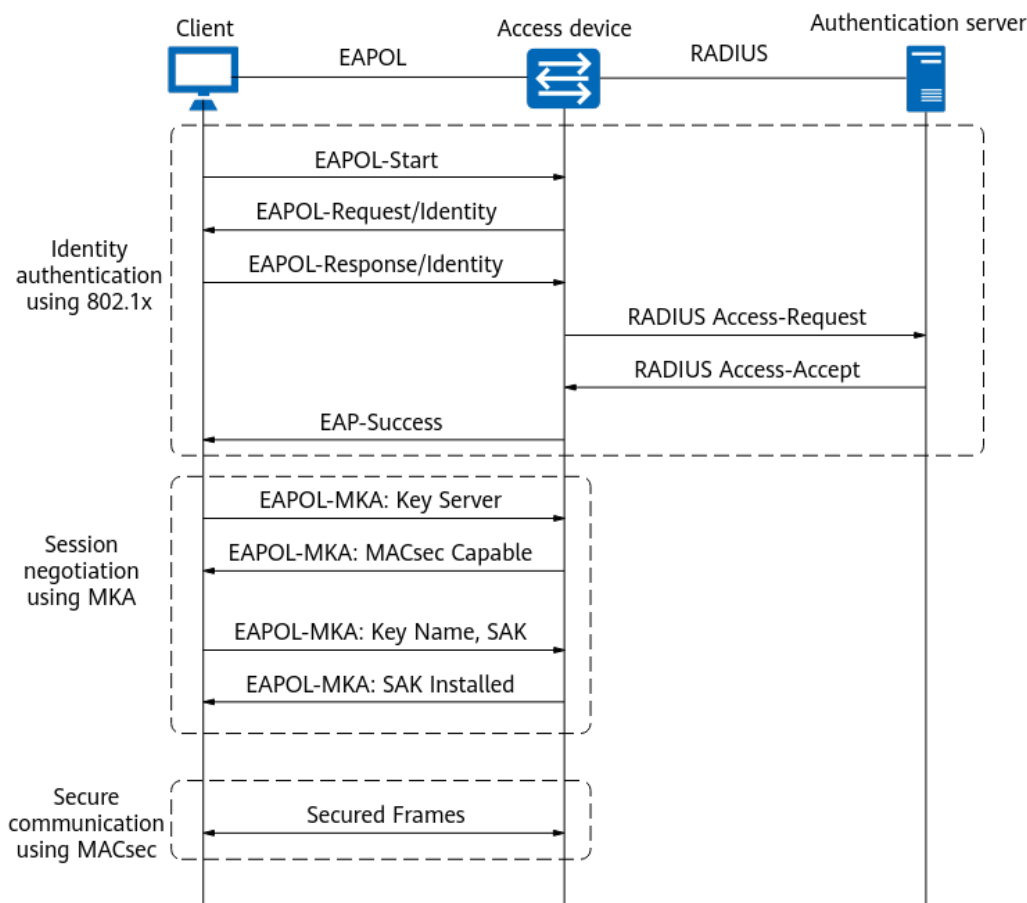
MACsec involves the following basic concepts:

- **Secure Connectivity Association (CA):** consists of two or more CA members that use the same key or cipher suite. A key possessed by a member of a given CA is called a CAK. CAKs are classified into two types: pairwise CAK and group CAK.
 - A pairwise CAK is used by two CA members comprising a CA.
 - A group CAK is used by three or more CA members comprising a CA.Currently, MACsec mainly applies to point-to-point networking and therefore uses the pairwise CAK. A pairwise CAK can be either the CAK generated during access control based on 802.1x or the pre-shared key (PSK). Huawei-issued base stations use the pairwise CAK generated during access control based on 802.1x.
- **Secure association (SA):** is a set of security parameters used for establishing secure channels between CA members. It includes the cipher suite used for encryption and keys used for integrity check. Each secure channel comprises a succession of SAs, and each SA has a unique key called SAK. An SAK is derived from a CAK and is used for encrypting data transmitted in secure channels. MKA restricts the maximum number of packets that can be encrypted by an SAK. If the number of packets using an SAK for encryption exceeds the maximum number, the SAK will be updated. For example, on a link with a data rate of 10 Gbit/s, an SAK is updated at an interval of 300 seconds or longer.

4.1.2 MACsec Signaling Process

MACsec interaction consists of four phases: identity authentication, session negotiation, secure communication, and session termination, as shown in [Figure 4-1](#).

Figure 4-1 MACsec signaling process



Identity Authentication

Before a client establishes a secure session with an access device, the client must have its identity authenticated over a port on the access device. After the authentication, the RADIUS server distributes the generated CAK to the client and access device. On the base station side, identity authentication is based on EAP-TLS, either EAP-TLS1.2 or EAP-TLS1.3. For details, see RFC5216 and RFC9190.

MACsec for BBU interconnection supports identity authentication using Huawei initial certificates.

MACsec for eCPRI supports identity authentication using Huawei initial certificates and operators' certificates.

- When MACsec for eCPRI uses a Huawei initial certificate, the **SSL.RRUHWCERTAUSHW (LTE eNodeB, 5G gNodeB)** parameter must be set to **ON** so that an RF module can use the Huawei initial certificate to set up a connection with a BBU.

- When MACsec for eCPRI requires an operator's certificate, the **APPCERT.RRUCERTREQSW (LTE eNodeB, 5G gNodeB)** parameter must be set to **ON**. When MACsec for eCPRI is already using an operator's certificate, it is recommended that the **SSL.RRUHCERTAUTHSW (LTE eNodeB, 5G gNodeB)** parameter be set to **ON**. Otherwise, if the operator's certificate is abnormal, identity authentication cannot be performed using a Huawei initial certificate. As a result, MACsec link setup fails.

 NOTE

The **SSL.RRUHCERTAUTHSW (LTE eNodeB, 5G gNodeB)** parameter can be set only when the **SSL.RRUTLSCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **TLS** or **TLS_PLAINTEXT** and the **SSL.RRUAUTHMODE (LTE eNodeB, 5G gNodeB)** parameter is set to **PEER**.

Session Negotiation

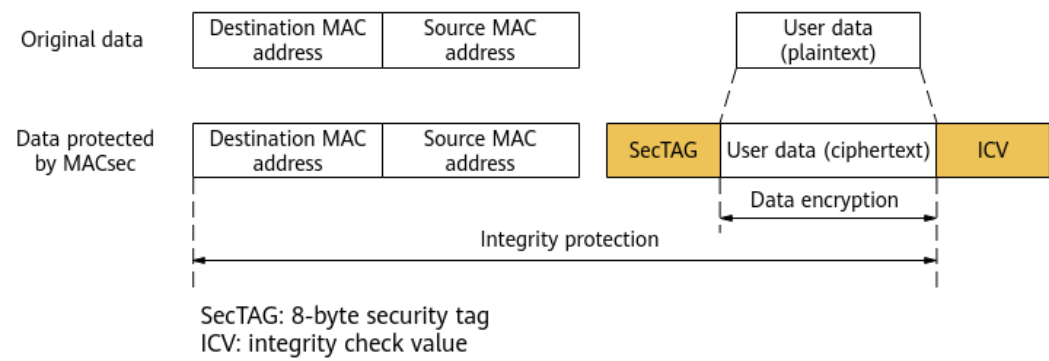
With the CAK, the client starts a negotiation session with the access device using EAPOL-MKA packets. During the negotiation session, the client and access device both use the MKA to inform each other of their capabilities and required parameters (such as priorities and whether the session needs to be encrypted). The access device is automatically selected as the key server in this process. The key server derives an SAK for encrypting data packets from the CAK and distributes the SAK to the client.

Secure Communication

After the negotiation session is complete, both the client and access device have the SAK and use it for encrypting data packets. Then, they start secure communication. MACsec measures include data encryption and integrity check. The anti-replay and perfect forward secrecy (PFS) functions are now not implemented on base stations because there is a low probability of replay attacks and key cracking in both BBU interconnection and eCPRI scenarios.

- **Data encryption**
After a MACsec negotiation succeeds, the sender encrypts the data and the receiver then decrypts the data. Keys used for encryption and decryption are obtained during MKA negotiation.
- **Integrity protection**
In a MACsec-encapsulated data frame, a key derived from the CAK is used to calculate the integrity check value (ICV) and the value is appended to the end of the MACsec packet. [Figure 4-2](#) shows the MACsec packet encapsulation.

Figure 4-2 MACsec packet encapsulation



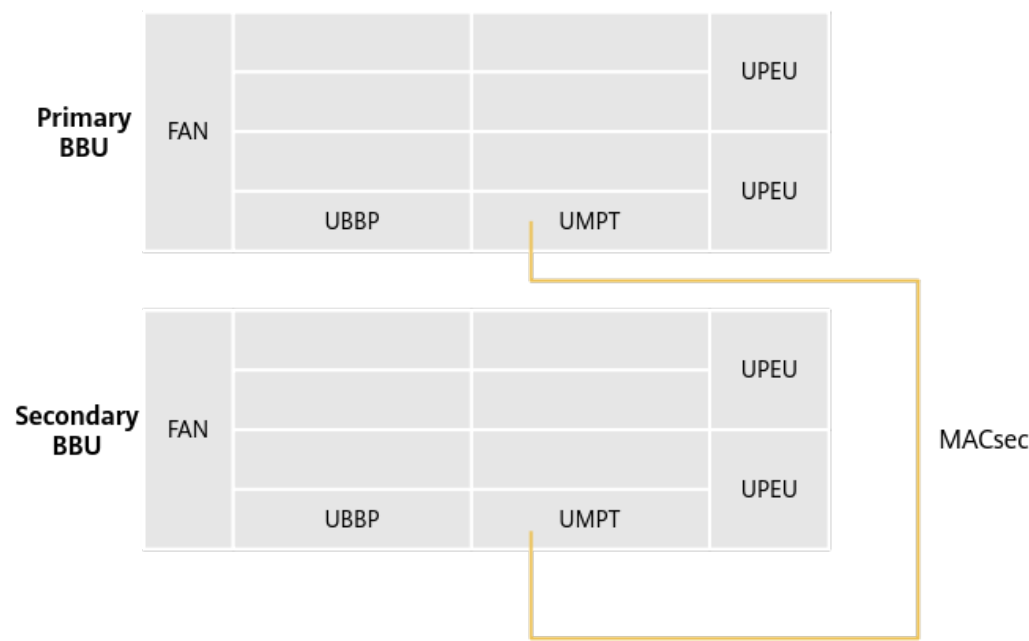
When the access device receives the MACsec packet, it also uses the key obtained during MKA negotiation to calculate the ICV and compares the result with the ICV carried in the packet. If the two ICVs are the same, the packet is valid. Otherwise, the packet is discarded.

MACsec supports the GCM-AES-128 cipher suite for data encryption and integrity protection of base stations.

4.1.3 Application in Base Stations

For BBU interconnection: MACsec can be used to protect data transmitted between UMPT boards in interconnected BBUs, as shown in Figure 4-3.

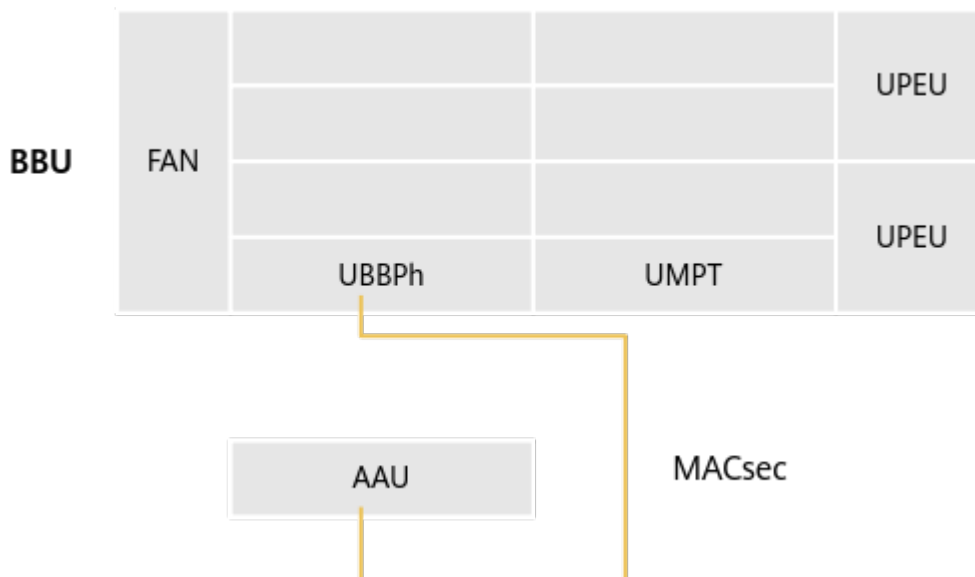
Figure 4-3 BBUs interconnected using the UMPT boards



For eCPRI: MACsec can be used to protect eCPRI data between the BBU and AAU, as shown in Figure 4-4. eCPRI data includes control-plane data, user-plane data, and synchronization-plane data. The **CPRIPORT.MACSECCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is involved.

- If this parameter is set to **PLAINTEXT**, only plaintext messages at the MAC layer can be used.
- If this parameter is set to **MACSEC**, only the MACsec protocol can be used.
- If this parameter is set to **MACSEC_PLAINTEXT**, either plaintext messages at the MAC layer or the MACsec protocol can be used.

Figure 4-4 A BBU and an AAU connected using the eCPRI protocol



NOTE

In eCPRI scenarios, MACsec can protect LLDP, TOPO, REC, and ARP packets only after the MACsec link setup succeeds.

4.2 Network Analysis

4.2.1 Benefits

This feature provides MAC-layer security protection for data transmitted through BBU interconnection or eCPRI ports and increases data transmission security.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

When MACsec is used for eCPRI, the values of the following indicators may increase:

- BBP CPU usage
VS.BBUBoard.CPULoad.Mean (LTE eNodeB, 5G gNodeB)
- Delay over the eCPRI link in the uplink and downlink
 - **VS.FronthaulLink.eCPRILink.UplinkDelayMeans (LTE eNodeB, 5G gNodeB)**

- **VS.FronthaulLink.eCPRILink.DownlinkDelayMeans** (LTE eNodeB, 5G gNodeB)
- Indicators related to eCPRI ports
 - **VS.ECPRIPORT.RxBytes** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.RxMaxSpeed** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.RxMeanSpeed** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.RxMinSpeed** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.TxBytes** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.TxMaxSpeed** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.TxMeanSpeed** (LTE eNodeB, 5G gNodeB)
 - **VS.ECPRIPORT.TxMinSpeed** (LTE eNodeB, 5G gNodeB)

Modifying the MACsec connection type (specified by the **CPRIPORT.MACSECCONNTYPE** (*LTE eNodeB, 5G gNodeB*) parameter) for eCPRI may cause a change between the plaintext at the MAC layer and the MACsec protocol. During the change, packet loss will occur, affecting services.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Public Key Infrastructure (PKI)	None	<i>PKI</i>	This feature is required if digital certificate authentication needs to be used for MACsec.

RAT	Function Name	Function Switch	Reference	Description
NR FDD NR TDD LTE FDD LTE TDD UMTS	eCPRI	None	<i>eCPRI</i>	This feature is required if digital certificate authentication needs to be used for MACsec.
NR FDD NR TDD	Security Mechanism (PKI)	None	<i>PKI</i>	This feature is required if digital certificate authentication needs to be used for MACsec.

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations
UMTS/LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For BBU interconnection:

Only UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series boards can be used.

For eCPRI:

Only UBBPh3a, UBBPh3b, and UBBPh3 boards can be used.

RF Modules

For BBU interconnection:

No requirements

For eCPRI:

Only AAUs introduced in SRAN20.1 and later versions support this function. For details about AAU models that support the eCPRI protocol, see CPRI port specifications in AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

The UMPT boards of the BBUs must be directly connected through network cables, without using any switching devices. Similarly, BBUs and eCPRI-compliant AAUs must be directly connected through optical fibers.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

This feature is activated by default and does not need to be configured in BBU interconnection scenarios. The following sections describe how to configure data in eCPRI scenarios.

4.4.2 Activation Verification

For BBU interconnection:

Run the **DSP CTRLNKSTAT** command on the primary or secondary BBU. If the value of the **MACSec Negotiation State** parameter is **Succeeded**, this feature works properly.

For eCPRI:

Run the **DSP CPRIPORT** command on the BBU. If the value of the **MACsec Connection Type** parameter is **MACSEC** and the value of the **MACsec Connection State** parameter is **NORMAL**, this feature works properly.

4.4.3 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

1. IEEE 802.1AE-2006
2. *SRAN Networking and Evolution Overview*